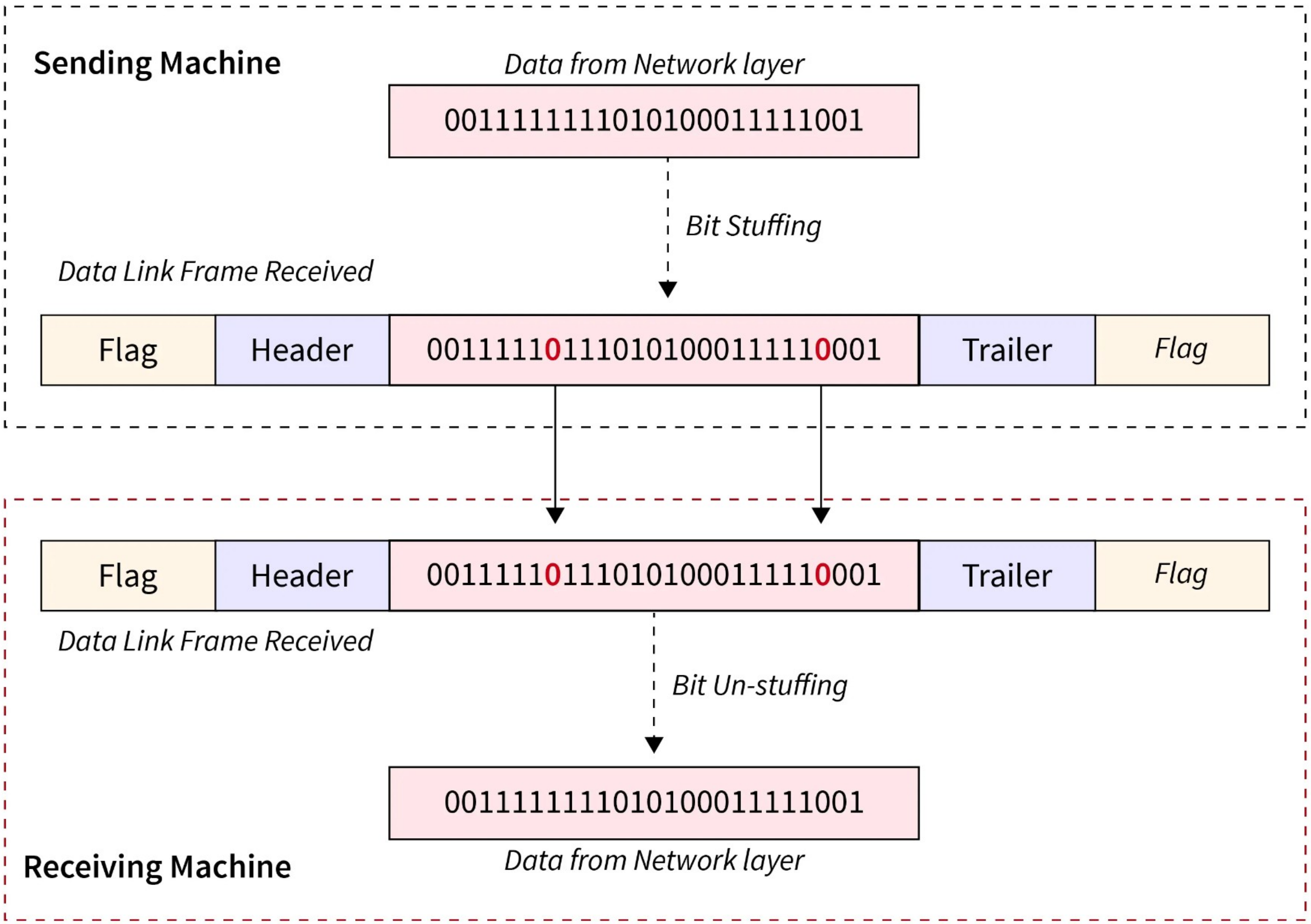
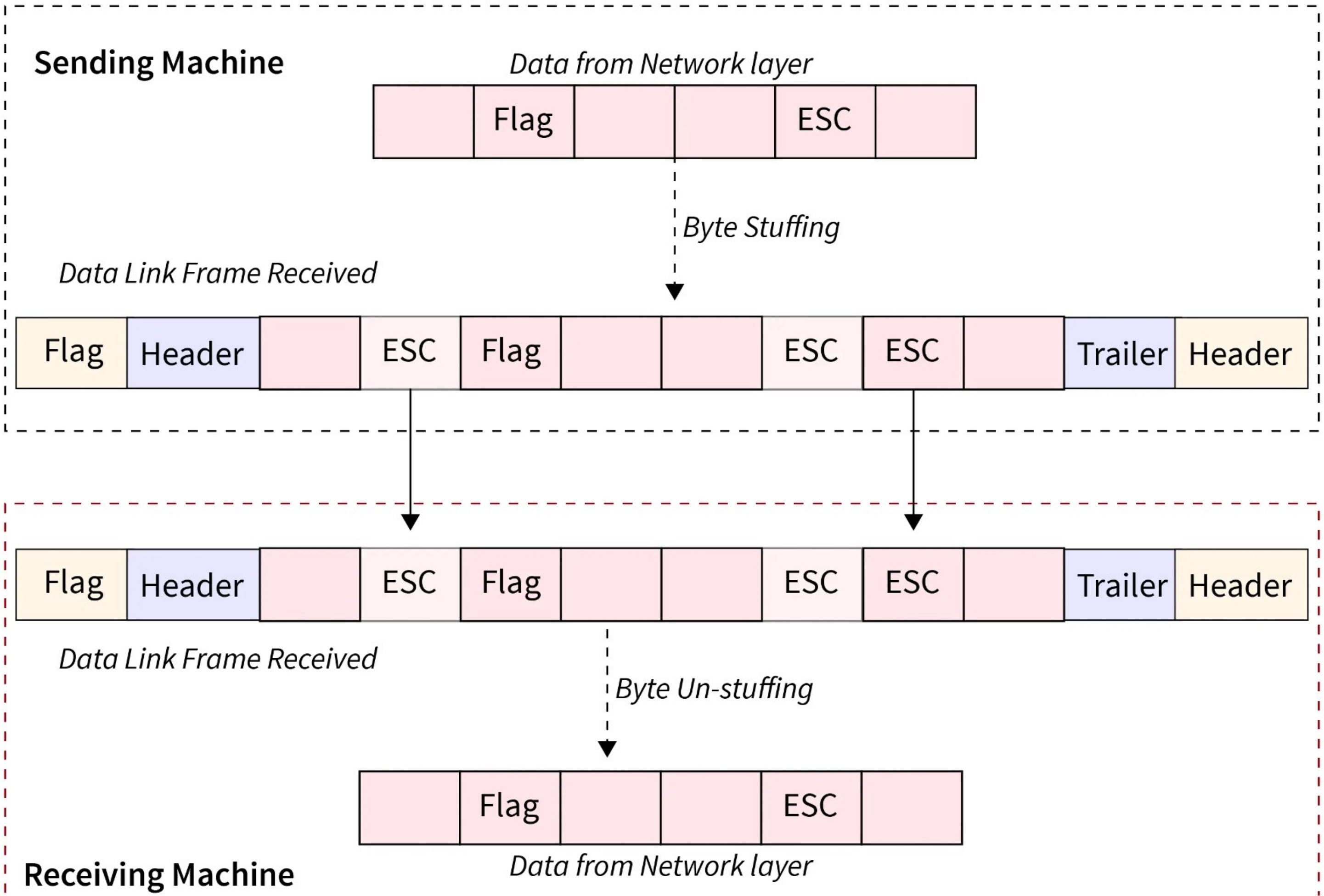
Framing

Bit Stuffing



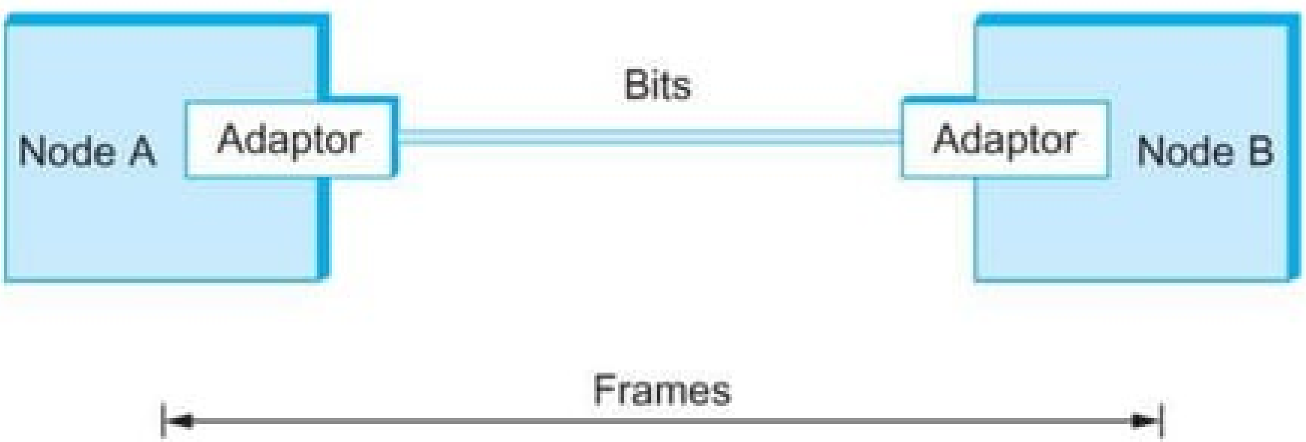
Byte Stuffing/Character Stuffing



Framing

Framing

- Data link layer translates a stream of bits from the physical layer into the larger aggregate (or) discrete unit called **frames**.
- The hardware present in the data link layer is network adaptor. It enables the nodes to exchange frames.

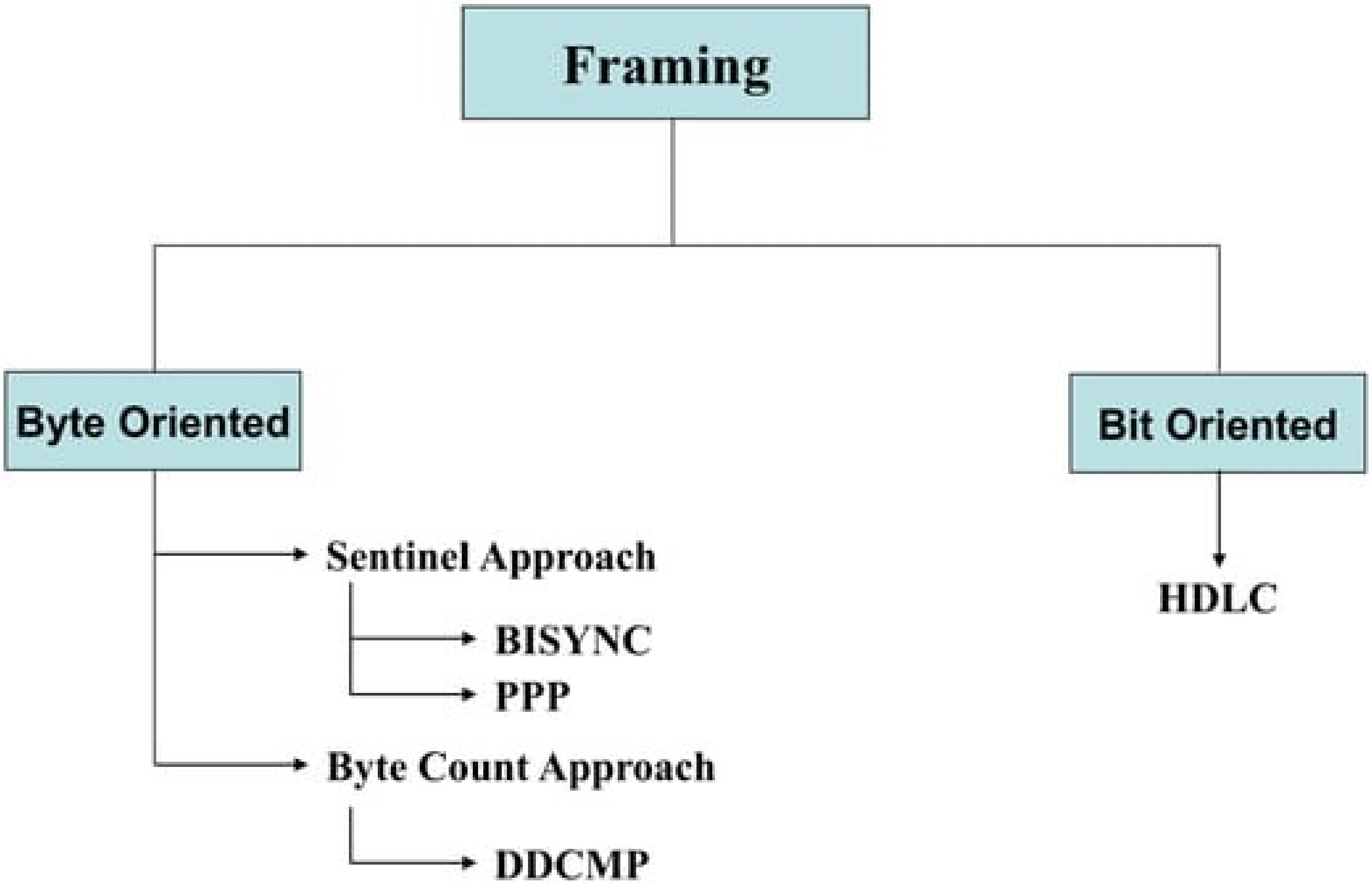


Framing

- When node A wishes to transmit a frame to node B, it tells its adaptor to transmit a frame from the node's memory. This results in a sequence of bits being sent over the link.
- The adaptor on node B then collects together the sequence of bits arriving on the link and deposits the corresponding frame in B's memory.
- Recognizing exactly what set of bits constitute a frame

  i.e., determining where the frame begins and ends is the issue faced by the adaptor

Framing


Framing
Byte Oriented
Bit Oriented
Sentinel Approach
BISYNC
PPP
Byte Count Approach
DDCMP
HDLC

Framing

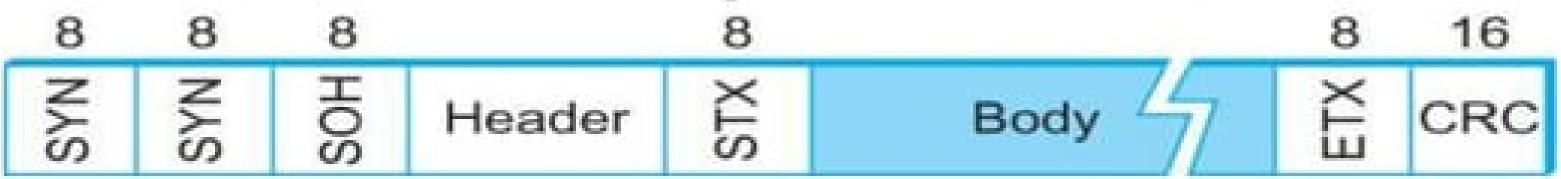
Byte Oriented Protocols:

– It views each frame as a **collection of bytes** (Character) rather than a collection of bits.

a) **Sentinel Approach:**

BISYNC (Binary Synchronous Communication)

The frame format of the BISYNC protocol is,



- Frames transmitted beginning with leftmost field
- **SYN** - Beginning of a frame is denoted by sending a special SYN (synchronize) character
- **STX & ETX** - Data portion of the frame is contained between special sentinel character STX (start of text) and ETX (end of text)
- **SOH (**Start of Header) – Header information is start from this sentinel character

- **CRC (Cyclic Redundancy Check)** – Used for error detection in the frame.

Problems in the sentinel approach,

- ETX character might appear in the data portion of the frame. So it leads to the wrong end of text calculation.
- BISYNC overcomes this problem by "escaping" the ETX character by preceding it with a DLE (data-link-escape) character whenever it appears in the body of a frame; the DLE character is also escaped (by preceding it with an extra DLE) in the frame body. This approach is called as "**character stuffing**".

PPP (Point to Point Protocol)

- Commonly used in dial up modem links.
- It also uses the "Character Stuffing"

PPP Frame format as follows,

- **Flag** - Special start of text character denoted as Flag
  - 0 1 1 1 1 1 1 0
- **Address, control** - default values
  - **Address – 11111111**
  - **Control – 11000000**
- **Protocol** - Used for demultiplexing. It identifies the high level protocol such as IP / IPX.
- **Payload –** Frame data (default size is 1500 bytes)
- **Checksum** - for error detection

Byte Stuffing in PPP,

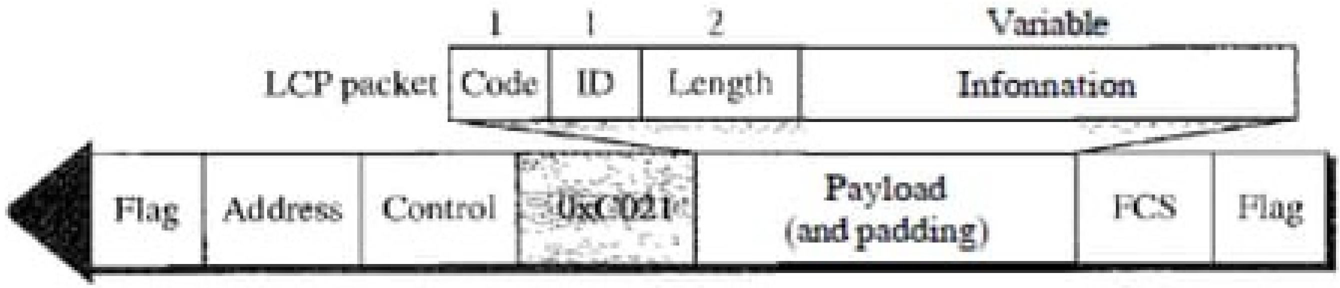
- Whenever the flag value appears in the data section of the frame, **escape byte (01111101)** stuffed into the data to tell the receiver that the next byte is not a flag.

Protocols used by PPP,

- LCP (Link Control Protocol)
- AP (Authentication Protocol)
- NCP (Network Control Protocol)

LCP (Link Control Protocol)

- Responsible for establishing, maintaining, configuring & terminating the links.
- All LCP packets are carried in the **payload of the PPP frame** with protocol field set to **"C021"** in hexadecimal.



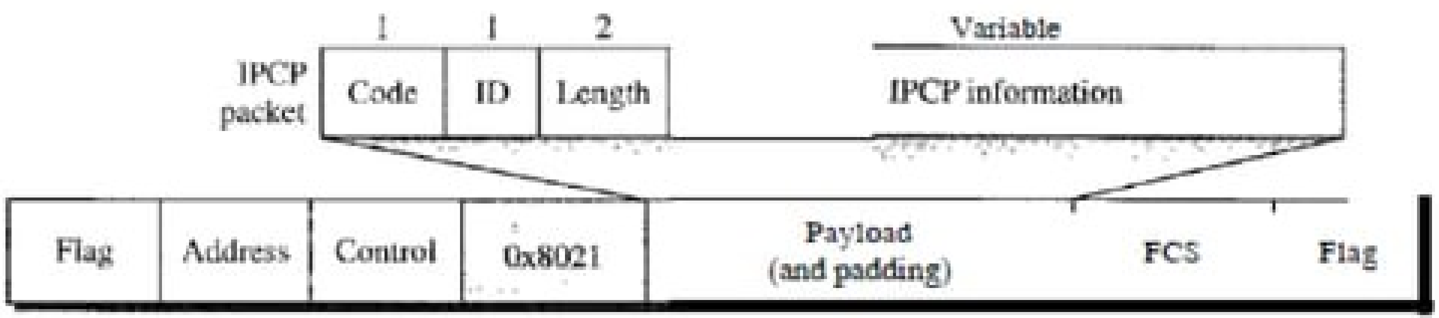
- **Code** – defines the type of the LCP packet. There are 11 types of LCP packet is available.
  - These 11 types of LCP packets are categorized into three category.
    - ➢ First category – comprising the first four packet types **(o×01 - o×04)** used for link configuration during link establishment.
    - ➢ Second Category – comprising packet types o×05, o×06 for the termination of the link.
    - ➢ Third Category – comprising last five packet types (o×07 - o×11) used for link monitoring and debugging.
- **Id –** value that matches the request and reply.
- **Length** – defines the total length of the LCP packet.
- **Information** – optional information.

AP (Authentication Protocol)

- PPP is mainly designed for dial up links, where the user identity verification is necessary
- **Authentication** – validating the identity of a user.
- PPP uses two authentication protocols,
    - PAP (Password Authentication Protocol)
    - CHAP (Challenge Handshake Authentication Protocol)

NCP (Network Control Protocol)

- IPCP (Internet Protocol Control Protocol) is used as a network control protocol.



- IPCP defines 7 types of packets distinguished by the code values.

b) Byte Count Approach:

DDCMP (Digital Data Communication Message protocol)

- DDCMP uses the count field that contains the no. of bytes in the frame to find the end of the frame.
- Suppose, if count field is corrupted framing error will arise.

SYN – starting of the frame.
count – no. of bytes in the frame.

Bit Oriented Protocols

– It simply views the frame as collection of bits.

HDLC

- Earlier called as SDLC (Synchronous Data Link Control)

> ➢ **Beginning and Ending Sequence** – denotes both the starting & end of the frame with the bit sequence 01111110.
>
> - This sequence is also transmitted during any times that the link is idle, so that the sender and receiver can keep their clock synchronization.

- **Bit Stuffing**

Sending side:

❖ On the sending side, any time five consecutive 1's have been transmitted from the body of the message (i.e. excluding when the sender is trying to send the distinguished 01111110 sequence)

- The sender inserts '0' before transmitting the next bit.

Receiving side:

- On the receiving side,
  - Five consecutive 1's
    - Next bit **0** - Stuffed, so discard it
      1 - Either End of the frame marker (or) Error has been introduced in the bit stream
    - Look at the next bit

      If 0 (01111110) → End of the frame marker

      If 1 (01111111) → Error, discard the whole frame
    - The receiver needs to wait for next 01111110 before it can start receiving again.